DAY 5 · FRIDAY 20 NOVEMBER 2026

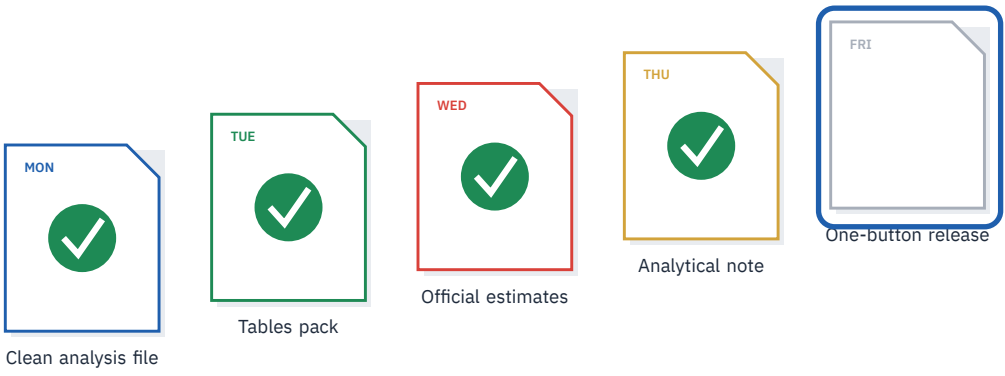
One button for the whole release

Macros, loops, a master do-file and an automated report



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Five days, five products



YOUR TURN · TASK 1

Today’s files

DO

Type:

```
dir "C:/NBS_Stata/day5"
```

YOU SHOULD SEE

The day’s files: the daily check and the folders `catchup`, `labs`, `modules`, `practice`, `refactor`.

IF NOT

Nothing listed? Copy the `day5` folder from the USB stick into `C:\NBS_Stata`.

Yesterday’s check: the results are on the board. The unclear points open this morning.

FRIDAY MORNING AT NBS

Next quarter you are on leave. Who produces the release?

WHAT WE HAVE

Four do-files from this week, each run by hand, in the right order, on your laptop.

WHAT WE NEED

One file that a colleague opens, changes one line in, and runs.

WHY IT IS HARD

Paths, settings and the order of steps live in your head, not in the files.

By this afternoon: one command, from the raw file to a Word release.

This afternoon one command will rebuild everything you produced this week.

What you will be able to do today

AFTER THIS PART YOU CAN

- keep settings and lists in macros, and build every path from one root
- repeat a table for every region with a loop
- stop a do-file early, with a clear message, when an input is missing
- run a whole production from one master do-file
- write a Word release in which no number is typed by hand



1

DAY 5 · PART 1

Names for things you use again

THE PROBLEM

The path C:/NBS_Stata/data/ is typed forty times this week

WHAT WE HAVE

Do-files full of the same folder, the same quarter, the same list of variables.

WHAT WE NEED

Type each of them once, at the top, and reuse it everywhere.

WHY IT MATTERS

A colleague with another folder has to find and change every copy. One missed copy breaks the run.

A macro is a name for a piece of text.

Part 1: macros

AFTER THIS PART YOU CAN

- hold a list in a `local` and use it
- hold a setting in a `global` and build paths from it
- stamp outputs with `c(current_date)` and `c(stata_version)`
- say when a macro is text and when it is a number

You met `local` on Tuesday. Today it does the heavy lifting.

A list under one name:



Use it as `` `vars' ``: the left quote is the key left of 1, the right quote the key left of Enter.

YOUR TURN · TASK 2

A list, then a loop over it

DO

Start `day5/my_day5.do` with the usual top lines, then type:

```
use "data/lfs_teach.dta", clear
local vars "ilo_sex agegrp region"
display "`vars'"
foreach v of local vars {
    tabulate `v'
}
```

YOU SHOULD SEE

The three names, then three tables: sex, age group, region.

IF NOT

Nothing displayed? Run all lines together: a local lives only while the lines run.

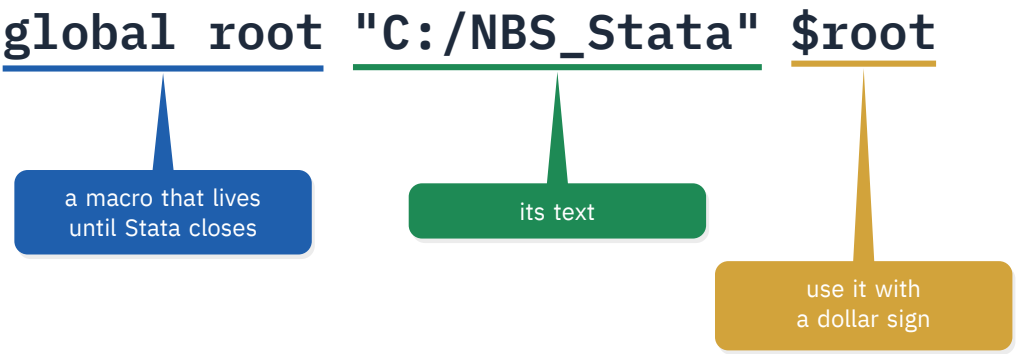
Read the list at work

```
. local vars "ilo_sex agegrp region"
. display "`vars'"
ilo_sex agegrp region (1)
. foreach v of local vars {
  2.     tabulate `v' (2)
  3. }

        Sex |      Freq.     Percent        Cum.
------------+----------------------------------- (3)
   1 - Male |      2,796       47.49       47.49
 2 - Female |      3,092       52.51      100.00
------------+-----------------------------------
      Total |      5,888      100.00
```

1. the local holds the text
2. the loop puts each name in turn into `` `v' ``
3. the first of three tables; age group and region follow

A setting for the whole session:



Local or global?

LOCAL

Lives while one do-file (or one selection of lines) runs, then disappears.

Use: `` `name' ``

For: lists, counters, numbers inside one file.

GLOBAL

Lives until Stata closes, in every do-file you run.

Use: `$name`

For: the few settings every file needs, such as the project folder and the quarter.

A global that is set by mistake in one file changes what the next file does.

RULE

Globals only for settings.

The project root, the quarter, one or two definitions. Everything else is a local.

Set them in one block, at the top of the master do-file.

YOUR TURN · TASK 3

Two settings

DO

Type:

```
global root "C:/NBS_Stata"
global quarter "2026Q1"
display "$root"
display "$quarter"
```

YOU SHOULD SEE

The folder and the quarter, each on its own line.

IF NOT

Read the settings

```
. global root "C:/NBS_Stata"

. global quarter "2026Q1"

. display "$root"
C:/NBS_Stata (1)

. display "$quarter"
2026Q1 (2)
```

1. `$root` is replaced by its text before the command runs
2. the same for `$quarter`

Every path starts from the root

```
use "$root/data/lfs_raw.dta", clear
save "$root/data/lfs_${quarter}_teach.dta", replace
log using "$root/logs/master_${quarter}.log", replace text
```

- a colleague with another folder changes one line: `global root`
- `${quarter}` with braces: the name ends where the brace ends
- next quarter, `2026Q2` names every file

YOUR TURN · TASK 4

Open the raw file from the root

DO

Type:

```
use "$root/data/lfs_raw.dta", clear
describe, short
```

YOU SHOULD SEE

The raw file opens: 5,895 observations.

IF NOT

File not found? Run Task 3 again: the global must be set first.

The path was built for you

```
. use "$root/data/lfs_raw.dta", clear
(NBS LFS Q1 2026 - workshop teaching extract (raw))

. describe, short

Contains data from C:/NBS_Stata/data/lfs_raw.dta
 Observations:         5,895                  NBS LFS Q1 2026 - workshop teaching extract (raw)
Sorted by: unique_ind
```

1. `$root/data/...` became `C:/NBS_Stata/data/...`
2. 5,895 rows, 90 variables: Monday's raw file

COMMON ERROR

A global with a typo is empty, silently

```
. display "Quarter: $quater"
Quarter:
. use "$rot/data/lfs_raw.dta", clear
file /data/lfs_raw.dta not found
```

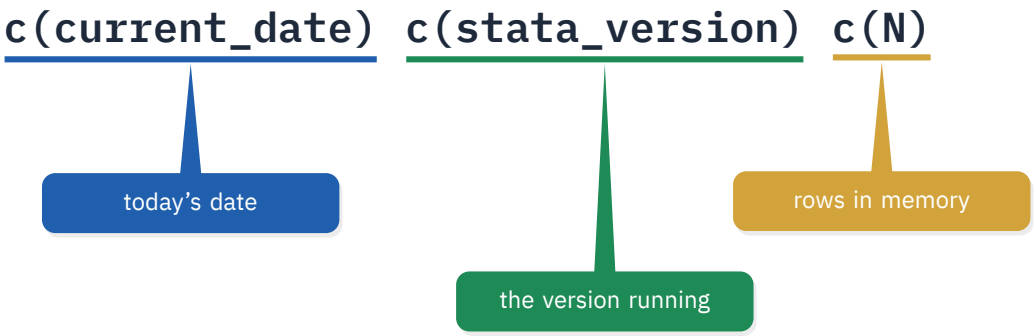
WHY

`$quater` and `$rot` were never set. Stata replaces them with nothing and carries on: the path becomes `/data/lfs_raw.dta`.

FIX

Check the spelling of the name in the error. `macro list` shows every global that exists.

Facts Stata knows:



YOUR TURN · TASK 5

Stamp a run

DO

Type:

```
display "Run on `c(current_date)' with Stata `c(stata_version)'"
display c(N)
```

YOU SHOULD SEE

One sentence with today's date and your version, then a number.

IF NOT

EXTRA What does `c(pwd)` show?

Read the stamp

```
. display "Run on `c(current_date)' with Stata `c(stata_version)'"
Run on 20 Oct 2026 with Stata 19.5 (1)

. display c(N)
5895 (2)

. display "`c(pwd)'"
C:\NBS_Stata (3)
```

1. the date and version, typed by nobody
2. rows in memory: the raw file
3. the working folder

YOUR TURN · TASK 6

A local lives for one run

DO

Type both lines and run them together:

```
local tmp "only here"
display "[`tmp']"
```

YOU SHOULD SEE

```
[only here].
```

IF NOT

EXTRA Now select only the `display` line and run it alone: the brackets are empty.

Inside one run

```
. local tmp "only here"

. display "[`tmp']"
[only here] 1
```

1. the local exists while these lines run, and only then

YOUR TURN · TASK 7

Which globals are set?

DO

Type:

```
macro list root quarter
```

YOU SHOULD SEE

Two lines: each name and its text.

IF NOT

EXTRA `macro list` alone lists all of them, including Stata's own.

Read the list

```
. macro list root quarter
root:           C:/NBS_Stata (1)
quarter:        2026Q1 (2)
```

1. the project root
2. the quarter

COMMON ERROR

A local called like a global

```
. local q "2026Q1"
. display "[$q]"
[]
```

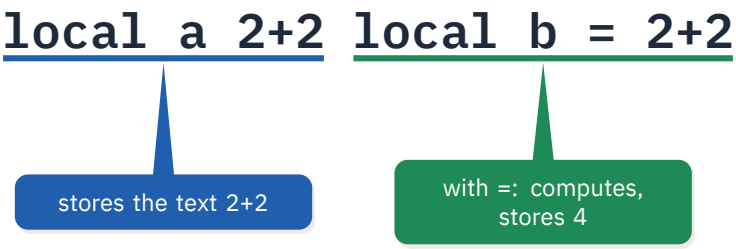
WHY

q is a local. $q asks for a global called q, which does not exist: empty, no error.

FIX

Locals with quotes, globals with a dollar: `q' gives 2026Q1.

Text, or a number?



YOUR TURN · TASK 8

The equals sign

DO

Type:

```
local a 2+2
local b = 2+2
display "`a'"
display "`b'"
display `a' * 3
display `b' * 3
```

YOU SHOULD SEE

Four lines: 2+2, 4, then two different numbers.

IF NOT

EXTRA Work out the last two before you run them.

Read the difference

```
. local a 2+2

. local b = 2+2

. display "`a'"
2+2

. display "`b'"
4

. display `a' * 3
8 1

. display `b' * 3
12 2
```

1. `` `a' * 3 `` becomes `2+2 * 3`: 8
2. `` `b' * 3 `` becomes `4 * 3`: 12

A range in one function

```
count if inrange(ilo_age, 15, 24)
count if ilo_age >= 15 & ilo_age <= 24
```

- `inrange(x, a, b)`: true when x is between a and b, both included
- the same count as the second line, which you know from Day 1
- a missing x is never in range

YOUR TURN · TASK 9

A definition as a setting

DO

Type:

```
use "$root/data/lfs_teach.dta", clear
global youth_max 24
count if inrange(ilo_age, 15, $youth_max)
```

YOU SHOULD SEE

The number of persons aged 15 to 24.

IF NOT

EXTRA Set `youth_max` to 29 and count again.

One setting, two definitions of youth

```
. global youth_max 24

. count if inrange(ilo_age, 15, $youth_max)
  720

. global youth_max 29

. count if inrange(ilo_age, 15, $youth_max)
  1,101

. global youth_max 24
```

1. 720 aged 15-24, then 1,101 aged 15-29: only the global changed

ON YOUR OWN · A

The teaching file, from the root

THE QUESTION

Open `lfs_teach.dta` with a path built from `$root`. How many rows and how many variables does it have?

HINT

`c(N)` and `c(k)`.

ON YOUR OWN · A · ANSWER

```
use "$root/data/lfs_teach.dta", clear
display c(N)
display c(k)
```

```
. display c(N)
5888

. display c(k)
92
```

1

1 5,888 persons and 92 variables: Monday's cleaned file

ON YOUR OWN · B

Young people in the labour force

THE QUESTION

With `youth_max` set to 29, how many persons aged 15 to 29 are in the labour force (`ilo_lfs19` 1 or 2)? And with 24?

HINT

`inrange()` and `inlist()` in one `count if`.

ON YOUR OWN · B · ANSWER

```
global youth_max 29
count if inrange(ilo_age, 15, $youth_max) ///
& inlist(ilo_lfs19, 1, 2)
```

```
. global youth_max 29

. count if inrange(ilo_age, 15, $youth_max) & inlist(ilo_lfs19, 1, 2)
  697 (1)

. global youth_max 24

. count if inrange(ilo_age, 15, $youth_max) & inlist(ilo_lfs19, 1, 2)
  361

.
. * ============================ PART 2  LOOPS ============================
```

1. 697 aged 15-29 in the labour force; 361 aged 15-24

QUICK CHECK

A do-file uses `$quater` by mistake. What happens?

A. red error: global not found

B. it is replaced by nothing and the line runs on

C. Stata asks for a value

ANSWER

A do-file uses `$quater` by mistake. What happens?

A. red error: global not found

B. it is replaced by nothing and the line runs on

C. Stata asks for a value

Log M04: ”Quarter:” with nothing after it, and the path /data/lfs_raw.dta.

Part 1 on one page

CHEAT CARD · KEEP THIS ONE

`local name "text"`	a macro for this run
`` `name' ``	use a local
`global name "text"`	a setting for the session
`$name ${name}`	use a global
`local n = 2+2`	with =: the result
`c(current_date) c(N)`	Stata's own facts
`macro list`	every macro that is set

Practice on your own

PRACTICE SET · practice_D5.do, PART 1

CORE	P1	Locals `lo` and `hi`; count persons aged lo to hi.
CORE	P2	A local `wk = 48`: employed with usual hours above it.
CORE	P3	How many variables in the teaching file? (`c(k)`)
EXTENSION	P4	Open the raw file from `$root`; type `c(N)`.
CHALLENGE	P5	What does `display `a' * 3` show when a holds 2+2?

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

Your settings now live in one place, and every path is built from them.

lost? open **catchup_D5_S1.do**



2

DAY 5 · PART 2

Seven regions, one block of code

THE PROBLEM

The same estimate for each of the seven regions

WHAT WE HAVE

One line that gives the unemployment rate of one region.

WHAT WE NEED

The same line seven times, with the region number changed each time.

WHY NOT COPY

Seven copies, seven chances for a typo, seven places to correct next quarter.

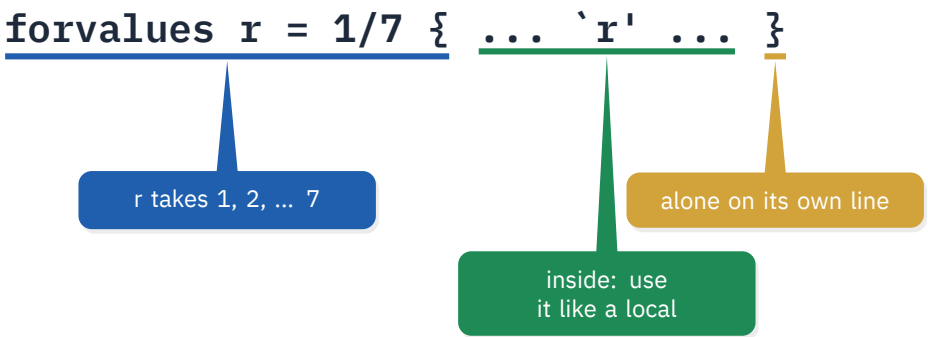
A loop writes the copies for you.

Part 2: loops and checks

AFTER THIS PART YOU CAN

- repeat a block for 1 to 7 with `forvalues`
- repeat over a list of numbers, variables or words with `foreach`
- take a region’s name from its value label
- check a command with `capture` and `_rc`, and stop with a message

Count from 1 to 7:



What the loop writes for you

```
forvalues r = 1/7 {
  svy, subpop(if lf == 1 & region == `r'): mean unemp
}
```




```
svy, subpop(if lf == 1 & region == 1): mean unemp
svy, subpop(if lf == 1 & region == 2): mean unemp
svy, subpop(if lf == 1 & region == 3): mean unemp
…
svy, subpop(if lf == 1 & region == 7): mean unemp
```

YOUR TURN · TASK 10

One rate per region

DO

Type:

```
use "$root/data/lfs_svy.dta", clear
quietly generate lf = (ilo_lfs19 == 1 | ilo_lfs19
== 2)
quietly generate unemp = (ilo_lfs19 == 2)
forvalues r = 1/7 {
    quietly svy, subpop(if lf == 1 & region ==
`r'): mean unemp
    matrix T = r(table)
    display "Region `r': " %5.2f 100 * T[1, 1]
}
```

YOU SHOULD SEE

Seven lines: Region 1 to Region 7 with a rate in percent.

IF NOT

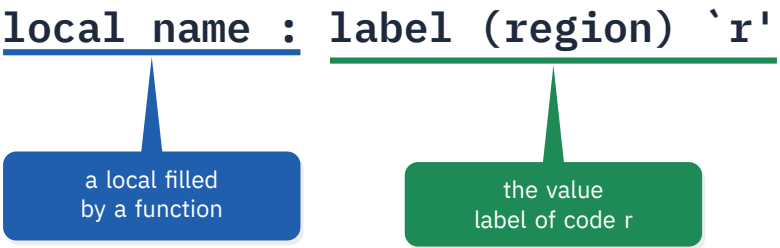
Red text about the brace? It must be the last thing on the first line.

Seven estimates, one block

```
. forvalues r = 1/7 {
  2.     quietly svy, subpop(if lf == 1 & region == `r'): mean unemp
  3.     matrix T = r(table)
  4.     display "Region `r': " %5.2f 100 * T[1, 1]
  5. }
Region 1:  3.21
Region 2:  4.53
Region 3:  3.62
Region 4:  2.68
Region 5:  5.32
Region 6:  0.83
Region 7:  1.63
```

1. Region 1, North: 3.21%
2. Region 6: 0.83%, the lowest
3. the last pass: r = 7

The region's name, not its number:



YOUR TURN · TASK 11

Names in the loop

DO

Type:

```
forvalues r = 1/7 {
    local name : label (region) `r'
    quietly svy, subpop(if lf == 1 & region ==
`r'): mean unemp
    matrix T = r(table)
    display "`name'" _col(24) %5.2f 100 * T[1, 1]
}
```

YOU SHOULD SEE

The same seven rates, now with the region names, in a column.

IF NOT

A table a reader can use

```
1- North                 3.21 (1)
2- Central 1             4.53
3- Central 2             3.62
4- EastSouth             2.68
5- West                  5.32
6- Praslin & LaDigue     0.83 (2)
7- Perseverance          1.63
```

(1) the label text, including its code: `1- North`

(2) Praslin & La Digue: 0.83%

It runs, but it is wrong: a variable without a value label

```
. local n : label (ilo_age) 30

. display "[`n']"
[30]
```

`ilo_age` has no value label, so `: label` returns the number itself, 30. No error, no warning. Check the variable has a label: `describe region` shows it in the value-label column.

YOUR TURN · TASK 12

The same rates in one command

DO

Type:

```
svy, subpop(lf): mean unemp, over(region)
```

YOU SHOULD SEE

One table with seven rows, a standard error and an interval each.

IF NOT

EXTRA Are the seven rates the same as the loop's?

Same rates, and one odd interval

```
. svy, subpop(lf): mean unemp, over(region)
------------------------------------------------------------------------------
                     |             Linearized
                     |       Mean   std. err.     [95% conf. interval]
---------------------+--------------------------------------------------------
       c.unemp@region |
            1- North |   .0320638   .0085409      .0151535    .0489741
        2- Central 1 |   .0453336   .0069975       .031479    .0591883
        3- Central 2 |   .0361783   .0110366      .0143267      .05803
        4- EastSouth |   .0267923   .0061746      .0145671    .0390175
             5- West |   .0531623   .0143954      .0246605    .0816642
6- Praslin & LaDigue |   .0083458   .0042431     -.0000553    .0167469
     7- Perseverance |   .0162846   .0077196      .0010003    .0315688
------------------------------------------------------------------------------
```

1. North 3.21%: the loop gave the same

2. Praslin: lower limit −.0000553, a negative rate. Part 5 fixes this in the release

Loop over a list:

```
foreach a of numlist 15 25 {
```

any numbers

```
foreach v of varlist x y {
```

variable names,
checked to exist

YOUR TURN · TASK 13

Over numbers

DO

Type:

```
foreach a of numlist 15 25 35 55 {
    quietly count if ilo_age >= `a'
    display "aged `a' and over: " r(N)
}
```

YOU SHOULD SEE

Four counts, falling.

IF NOT

Read the counts

```
aged 15 and over: 4627 (1)
aged 25 and over: 3907
aged 35 and over: 3118
aged 55 and over: 1463 (2)
```

1. 4,627 persons of working age
2. 1,463 aged 55 and over

YOUR TURN · TASK 14

Over variables

DO

Type:

```
foreach v of varlist ilo_sex agegrp {
    quietly svy, subpop(lf): mean unemp, over(`v')
    matrix T = r(table)
    display "`v': " %5.2f 100 * T[1, 1] "  " %5.2f
100 * T[1, 2]
}
```

YOU SHOULD SEE

Two lines: the first two rates by sex, then by age group.

IF NOT

Read the two lines

```
ilo_sex:  3.45   3.17 (1)
agegrp: 12.78   3.01 (2)
```

1. men 3.45%, women 3.17%
2. 15-24: 12.78%; 25-34: 3.01%

YOUR TURN · TASK 15

Steps other than one

DO

Type:

```
forvalues y = 2010(5)2025 {
    display "year `y'"
}
```

YOU SHOULD SEE

Four years.

IF NOT

EXTRA Try `1/7` with `(2)`: which regions?

Every fifth year

```
. forvalues y = 2010(5)2025 {
  2.     display "year `y'"
  3. }
year 2010
year 2015
year 2020
year 2025
```

1

1. 2010, 2015, 2020, 2025

COMMON ERROR

in instead of of

```
.     foreach v in varlist lf unemp {
  2.          summarize `v'
  3.      }
variable varlist not found
r(111);
```

WHY

`foreach v in` takes the words as they are: the first is the word `varlist`, which is not a variable.

FIX

`foreach v of varlist ...`: `of` plus the kind of list.

COMMON ERROR

Code on the line of the opening brace

```
.     forvalues r = 1/3 { display `r' }
program error:  code follows on the same line as open brace
r(198);
```

WHY

The brace must end its line. Stata does not run a loop body written on the same line.

FIX

Three lines: `forvalues ... {`, the body, `}`.

A loop inside a loop

```
forvalues r = 1/7 {
local name : label (region) `r'
forvalues s = 1/2 {
quietly svy, subpop(if lf == 1 & region == `r' & ilo_sex == `s'): mean unemp
matrix T = r(table)
display "`name'" _col(24) "sex `s'" _col(32) %5.2f 100 * T[1, 1]
}
}
```

- outer loop: 7 regions
- inner loop: 2 sexes for each region, 14 estimates
- indent each level: the braces stay readable

YOUR TURN · TASK 16

Region by sex

DO

Type the block on the left.

```
forvalues r = 1/7 {
    local name : label (region) `r'
    forvalues s = 1/2 {
        quietly svy, subpop(if lf == 1 & region ==
`r' ///
            & ilo_sex == `s'): mean unemp
        matrix T = r(table)
        display "`name'" _col(24) "sex `s'"
_col(32) %5.2f 100 * T[1, 1]
    }
}
```

EXTRA Write the results into a matrix R (Task after next).

YOU SHOULD SEE

Fourteen lines.

IF NOT

Fourteen rates

```
1- North                sex 1    4.92 (1)
1- North                sex 2    1.57
2- Central 1            sex 1    4.92
2- Central 1            sex 2    4.12
3- Central 2            sex 1    2.41
3- Central 2            sex 2    4.67
4- EastSouth            sex 1    2.69
4- EastSouth            sex 2    2.67
5- West                 sex 1    5.39
5- West                 sex 2    5.25
6- Praslin & LaDigue    sex 1    0.00 (2)
6- Praslin & LaDigue    sex 2    1.73
7- Perseverance         sex 1    0.82
7- Perseverance         sex 2    2.13
```

1. North: men 4.92%, women 1.57%
2. Praslin, men: 0.00. No unemployed man in the sample there

YOUR TURN · TASK 17

Into a matrix

DO

Type:

```
matrix R = J(7, 2, .)
forvalues r = 1/7 {
    forvalues s = 1/2 {
        quietly svy, subpop(if lf == 1 & region ==
`r' ///
            & ilo_sex == `s'): mean unemp
        matrix T = r(table)
        matrix R[`r', `s'] = 100 * T[1, 1]
    }
}
matrix colnames R = men women
matrix list R, format(%5.2f)
```

YOU SHOULD SEE

A 7 by 2 table: men, women.

IF NOT

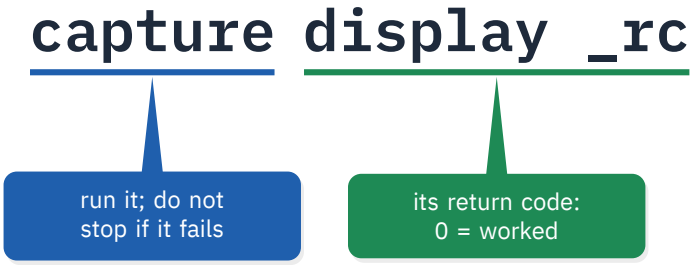
The same 14 numbers, kept

```
. matrix list R, format(%5.2f)

R[7,2]
      men   women  (1)
r1   4.92    1.57
r2   4.92    4.12
r3   2.41    4.67
r4   2.69    2.67
r5   5.39    5.25
r6   0.00    1.73
r7   0.82    2.13
```

(1) row 1 = North: ready for putexcel, as on Day 2

Did it work?


capture display _rc
run it; do not stop if it fails
its return code: 0 = worked

YOUR TURN · TASK 18

Two files, two return codes

DO

Type:

```
capture confirm file "$root/data/lfs_raw.dta"
display _rc
capture confirm file
"$root/data/lfs_2025Q4_raw.dta"
display _rc
```

YOU SHOULD SEE

0, then 601.

IF NOT

Read the codes

```
. capture confirm file "data/lfs_raw.dta"

. display _rc
0 (1)

. capture confirm file "data/lfs_2025Q4_raw.dta"

. display _rc
601 (2)
```

1. 0: the file is there
2. 601: file not found; the same code as the red message would show

Check the input, stop with a message

```
capture confirm file "$root/data/lfs_raw.dta"
if _rc {
display as error "Input missing: $root/data/lfs_raw.dta"
exit 601
}
```

- `if _rc`: true when the code is not 0
- `display as error`: a red line that names the file
- `exit 601`: stop the do-file here, with that code

YOUR TURN · TASK 19

Guard your do-file

DO

Add the five lines at the top of `my_day5.do`, after the settings. Run.

```
capture confirm file "$root/data/lfs_raw.dta"
if _rc {
    display as error "Input missing:
$root/data/lfs_raw.dta"
    exit 601
}
```

YOU SHOULD SEE

Nothing: the file is there, so the check is silent.

IF NOT

When the input is missing

```
. capture confirm file "$root/data/lfs_2025Q4_raw.dta"

. if _rc {
.     display as error "Input missing: $root/data/lfs_2025Q4_raw.dta"
Input missing: C:/NBS_Stata/data/lfs_2025Q4_raw.dta
.     exit 601
r(601);
```

1. the red line names the missing file
2. the file stops here: nothing below runs

COMMON ERROR

A subpopulation with nobody in it

```
. svy, subpop(if lf == 1 & region == 8): mean unemp
no observations in subpop() subpopulation
```

WHY

There is no region 8. A loop that runs past the last code meets an empty subpopulation.

FIX

Loop over the codes that exist: `1/7`. `tabulate region` shows them.

A production do-file checks its inputs before it starts, and stops with a clear message.

ON YOUR OWN · C

Participation, region by region

THE QUESTION

In a loop, give the design-based labour force participation rate of each region, with its name. Which region has the highest?

HINT

subpop: working-age persons of region r; mean of `lf`.

ON YOUR OWN · C · ANSWER

```
forvalues r = 1/7 {
local name : label (region) `r'
quietly svy, subpop(if wap == 1 & region == `r'):
mean lf
matrix T = r(table)
display "`name'" _col(24) %5.1f 100 * T[1, 1]
}
```

```
1- North                 71.3
2- Central 1             68.7
3- Central 2             58.6
4- EastSouth             65.3
5- West                  74.0
6- Praslin & LaDigue     68.9
7- Perseverance          67.0
```

1

1 West 74.0%, the highest; Central 2 58.6%, the lowest

ON YOUR OWN · D

Three flags in one loop

THE QUESTION

With one `foreach` over `lf unemp emp`, give the design-based share of each among the working-age population.

HINT

```
foreach v of varlist ...; subpop(wap).
```

ON YOUR OWN · D · ANSWER

```
foreach v of varlist lf unemp emp {
quietly svy, subpop(wap): mean `v'
matrix T = r(table)
display "`v'" _col(10) %6.2f 100 * T[1, 1]
}
```

```
lf        67.81
unemp      2.24
emp       65.57 1
```

1 participation 67.8%, employment-to-population 65.6%. 2.24% is unemployed per working-age person, not the unemployment rate (3.3%)

QUICK CHECK

After `capture confirm file "x.dta"`, `_rc` is 601. What does it mean?

A. the file was found

B. the file was not found

C. the file is open in another program

ANSWER

After `capture confirm file "x.dta"`, `_rc` is 601. What does it mean?

A the file was found

B the file was not found

C the file is open in another program

Log L09: 0 found, 601 not found.

Part 2 on one page

CHEAT CARD · KEEP THIS ONE

`forvalues r = 1/7 { ... }`	repeat for 1 to 7
`forvalues y = 2010(5)2025 {`	with a step
`foreach a of numlist 15 25 {`	over numbers
`foreach v of varlist x y {`	over variables
`local name : label (region) `r'`	the label of a code
`capture command`	run, do not stop on failure
`if _rc { display as error ... exit 601 }`	stop with a message

Practice on your own

PRACTICE SET · practice_D5.do, PART 2

CORE	P1	Count the labour force per region in a loop; type EastSouth.
CORE	P2	Design-based participation per region; type Praslin & La Digue.
CORE	P3	Employed aged 15, 25, 35, 55 and over; type the count for 55.
EXTENSION	P4	One `foreach` over lf unemp emp; type the share employed.
CHALLENGE	P5	Participation, region by sex; type women in West.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You can repeat any NBS table for every region and guard a do-file against missing inputs.

lost? open **catchup_D5_S2.do**



3

DAY 5 · PART 3

One file per job

THE PROBLEM

One do-file for the whole quarter grows to 800 lines

WHAT WE HAVE

Four days of do-files: cleaning, design, estimates, model. Each runs on its own.

WHAT WE NEED

Steps that can be run, checked and corrected one at a time, in a fixed order.

WHY NOT ONE FILE

An error on line 640 means rerunning everything to find it. Nobody can reuse half of it.

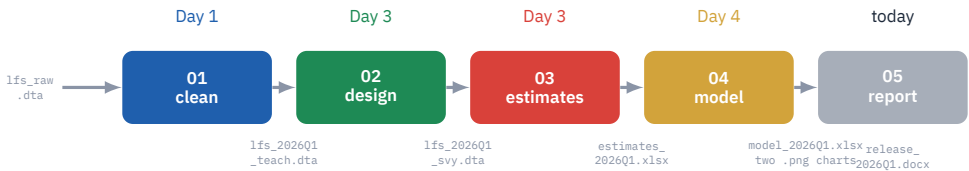
One module per job; a named file passes from one to the next.

Part 3: modules

AFTER THIS PART YOU CAN

- read the header of a module: what it reads, what it writes
- run one module from another file with `do`
- give a module the settings it needs
- say why a module must never rely on what is in memory

The week as an assembly line



Each module opens the file the one before it saved, and saves one file for the next.

The same stations as GSBPM

- **5 Process**: 01 cleans and codes, 02 declares the design and derives the flags
- **6 Analyse**: 03 produces the official estimates, 04 the model and the price series
- **7 Disseminate**: 05 writes the release

Source: UNECE, Generic Statistical Business Process Model, version 5.1 (2019).

YOUR TURN · TASK 20

The five modules

DO

Type:

```
local files : dir "$root/day5/modules" files
"*.do"
foreach f of local files {
    display "`f'"
}
```

YOU SHOULD SEE

Five file names, 01 to 05.

IF NOT

Nothing? Check the global root is set (Part 1).

Five modules

```
01_clean.do (1)
02_design.do
03_estimates.do
04_model.do
05_report.do
```

(1) numbered in the order they run: the names sort by themselves

A module begins with its header

```
* ====================================================================
* 01_clean.do  —  module 1: raw LFS file to the analysis file (Day 1)
* --------------------------------------------------------------------
* Course     : NBS Seychelles Stata training, Day 5 (20 Nov 2026)
* Author     : Endri Raco
* Version    : Stata 17 or newer (written with version 17)
* Inputs     : $root/data/lfs_raw.dta
* Outputs    : $root/data/lfs_${quarter}_teach.dta
* Packages   : none
* Called by  : master.do (needs the globals root and quarter)
* ====================================================================
version 17
```

Inputs and Outputs: what it reads and writes. Called by: the globals it needs.

Then Monday’s work, in one tidy file

```
* One person, one row: duplicates first, so later counts are of persons.
duplicates drop
isid unique_ind
isid unique_hhd ppno

* Don't-know (.a) and refused (.b) codes, as read from the labels.
mvdecode dem_mrt dem_edl, mv(97=.a \ 99=.b)
mvdecode mie_socpro, mv(97=.a)
mvdecode wkt_mj_acthrs, mv(997=.a)
assert missing(ilo_lfs19) if ilo_age < 15

* Teaching names for the design variables.
rename region_q region
rename district_q district
```

Every line was typed on Day 1. Here it is in order, with `$root` and `$quarter` in the paths.

What each module reads and writes

module	reads	writes
01_clean	lfs_raw.dta	lfs_2026Q1_teach.dta
02_design	lfs_2026Q1_teach.dta	lfs_2026Q1_svy.dta
03_estimates	lfs_2026Q1_svy.dta	estimates_2026Q1.xlsx
04_model	lfs_2026Q1_svy.dta, cpi_divisions_long.dta	model_2026Q1.xlsx, margins and infl .png
05_report	the two .xlsx, the two .png	release_2026Q1.docx

Each output name carries the quarter. Nothing passes from one module to the next except these files.

Run one do-file from another:

```
do "$root/day5/modules/01_clean.do"
```

run a do-file

its full path,
from the root

YOUR TURN · TASK 21

Look at module 01

DO

Open `day5/modules/01_clean.do` in the Do-file Editor (File > Open). Read the header. Do not run it yet.

YOU SHOULD SEE

The header, then the steps of Day 1.

IF NOT

EXTRA Which globals does it need?

YOUR TURN · TASK 22

Run module 01, with its settings

DO

In `my_day5.do`, type:

```
global root "C:/NBS_Stata"
global quarter "2026Q1"
global youth_max 24

do "$root/day5/modules/01_clean.do"
```

YOU SHOULD SEE

Module 01 runs line by line and ends with `file ... lfs_2026Q1_teach.dta` saved.

IF NOT

Red text? Read the first red line: usually a global not set.

The module runs, and saves

```
. do "$root/day5/modules/01_clean.do"
. duplicates drop
   (1)
Duplicates in terms of all variables

(7 observations deleted)
. save "$root/data/lfs_${quarter}_teach.dta", replace
(file C:/NBS_Stata/data/lfs_2026Q1_teach.dta not found) (2)
file C:/NBS_Stata/data/lfs_2026Q1_teach.dta saved
```

1. Day 1's duplicates: 7 rows removed
2. the file for module 02, named with the quarter

RULE

A module opens its own input.

It starts with `use`, never with whatever another file left in memory.

Then it runs the same alone, or inside the master.

YOUR TURN · TASK 23

Module 02 reads what 01 saved

DO

Type:

```
do "$root/day5/modules/02_design.do"
count if lf_youth == 1
```

YOU SHOULD SEE

The design is declared, six flags are built, the file is saved; then a count.

IF NOT

The flags, built from a setting

```
. count if lf_youth == 1
  361 (1)
```

(1) 361 young people in the labour force, with `youth_max` at 24

YOUR TURN · TASK 24

Module 03 and its Excel file

DO

Type:

```
do "$root/day5/modules/03_estimates.do"
import excel using
"$root/output/estimates_${quarter}.xlsx", ///
    sheet("National") firstrow clear
list
```

YOU SHOULD SEE

Ten design-based estimates scroll past; then the National sheet: three rows.

IF NOT

Excel open? Close it: putexcel cannot write to an open file.

The National sheet

```
    +-----------------------------------------------+
    | Indica~r        Rate       Lower       Upper |
    |-----------------------------------------------|
1.  |    unemp   3.3066046   2.6831721   4.0688347 |
2.  |   youth    12.776274   9.1196679   17.614851 |
3.  |    part    67.807314   65.811933   69.739783 |
    +-----------------------------------------------+
```

1. unemployment, youth unemployment, participation: rate, lower, upper, in percent

COMMON ERROR

A module run without its settings

```
. do "C:/NBS_Stata/day5/modules/01_clean.do"
. use "$root/data/lfs_raw.dta", clear
file /data/lfs_raw.dta not found
r(601);
```

WHY

The globals were dropped. `$root` is empty, so the path becomes `/data/lfs_raw.dta`.

FIX

Set the globals first, or run the module through the master, which sets them.

COMMON ERROR

A quarter with no files

```
. global quarter "2025Q4"

. use "$root/data/lfs_${quarter}_svy.dta", clear
file C:/NBS_Stata/data/lfs_2025Q4_svy.dta not found
r(601);
```

WHY

With `quarter` set to 2025Q4, module 03 looks for `lfs_2025Q4_svy.dta`, which modules 01 and 02 never made.

FIX

Run the modules in order for that quarter, or set the quarter back.

COMMON ERROR

A module that relies on memory

```
. clear
. svyset
no variables defined
```

WHY

After `clear` there is no data and no design. A module that starts with `svy:` instead of `use` fails like this.

FIX

First line of every module after the header: `use "$root/data/..." , clear`.

YOUR TURN · TASK EXTRA

Extra: your own Day 1

DO

Skippable. Replace the `do` line for module 01 by your own file:

```
do "$root/day1/my_day1.do"
```

YOU SHOULD SEE

Your Monday file runs; module 02 then needs `lfs_2026Q1_teach.dta`, which your file may not save.

IF NOT

EXTRA Make your file save that name, and module 02 accepts it.

ON YOUR OWN · E

Who writes, who reads?

THE QUESTION

Which module writes `lfs_2026Q1_svy.dta`, and which modules read it?

HINT

The Inputs and Outputs lines of the headers.

ON YOUR OWN · E · ANSWER

```
* 02_design.do
* Outputs    : $root/data/lfs_${quarter}_svy.dta
* 03_estimates.do
* Inputs     : $root/data/lfs_${quarter}_svy.dta
```

```
* Inputs     : $root/data/lfs_${quarter}_svy.dta,
$root/data/cpi_divisions_long.dta
```

1. 02 writes it; 03 and 04 read it

ON YOUR OWN · F

The youth rate in the file

THE QUESTION

After module 03, what youth unemployment rate does the National sheet hold, with its limits?

HINT

Row 2 of the sheet.

ON YOUR OWN · F · ANSWER

```
import excel using ///
"$root/output/estimates_${quarter}.xlsx", ///
sheet("National") firstrow clear
list
```

```
     +-----------------------------------------------+
     | Indica~r        Rate       Lower       Upper |
     |-----------------------------------------------|
  1. |    unemp    3.3066046   2.6831721   4.0688347 |
  2. |    youth    12.776274   9.1196679   17.614851 |
  3. |     part    67.807314   65.811933   69.739783 |
     +-----------------------------------------------+
```

1

1. 12.8%, from 9.1 to 17.6

QUICK CHECK

Module 03 is run on its own and fails at its first line. The most likely reason?

A. the globals root and quarter are not set

B. module 03 needs module 05 first

C. Excel is closed

ANSWER

Module 03 is run on its own and fails at its first line. The most likely reason?

A the globals root and quarter are not set

B module 03 needs module 05 first

C Excel is closed

Log D02: an empty $root turns the path into /data/....

Part 3 on one page

CHEAT CARD · KEEP THIS ONE

`do "$root/day5/modules/01_clean.do"`	run a module
`* Inputs : ... * Outputs : ...`	the header says what it reads and writes
`use "$root/data/..." , clear`	a module opens its own input
`save "$root/data/x_${quarter}.dta"`	and saves one named file
`local f : dir "folder" files "*.do"`	the files in a folder

Practice on your own

PRACTICE SET · practice_D5.do, PART 3

CORE	P1	Run module 01; type the number of persons it keeps.
CORE	P2	Run module 02; count the young labour force.
CORE	P3	Count the labour force aged 15 to 19, without saving.
EXTENSION	P4	Run module 03; type the youth rate from the National sheet.
CHALLENGE	P5	Module 03 with quarter 2025Q4, under `capture`: type the code.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

Your week is now five clean modules, each doing one job.

lost? open **catchup_D5_S3.do**



4

DAY 5 · PART 4

One file that runs them all

THE PROBLEM

Five modules, five settings, one right order

WHAT WE HAVE

Five modules that work when you run them by hand, in order, after setting three globals.

WHAT WE NEED

One file that does all of that, the same way every time.

WHY IT MATTERS

The order and the settings are the knowledge. Written down, anyone can run the quarter.

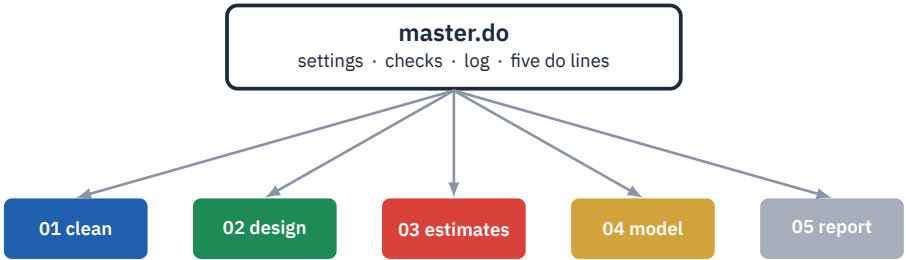
The master do-file is the release recipe.

Part 4: the master do-file

AFTER THIS PART YOU CAN

- write the settings block that a colleague changes
- check the inputs and create folders before anything runs
- keep one log for the whole run
- run the five modules and time the run

The master on top



run in this order, every time, by one key

master.do, top half

```
* ---- Settings: the only lines a colleague changes --------------------
version 17
clear all
global root "C:/NBS_Stata"
global quarter "2026Q1"
global youth_max 24

* ---- Checks: stop early, with a clear message ------------------------
foreach f in "data/lfs_raw.dta" "data/cpi_divisions_long.dta" {
capture confirm file "$root/`f'"
if _rc {
display as error "Input missing: $root/`f'"
exit 601
}
}
capture mkdir "$root/output"
capture mkdir "$root/logs"
```

master.do, bottom half

```
* ---- One log for the whole run ---------------------------------------
capture log close
log using "$root/logs/master_${quarter}.log", replace text
display "Run on `c(current_date)' at Stata `c(stata_version)', quarter $quarter"

* ---- The modules, in order -------------------------------------------
timer clear
timer on 1
do "$root/day5/modules/01_clean.do"
do "$root/day5/modules/02_design.do"
do "$root/day5/modules/03_estimates.do"
do "$root/day5/modules/04_model.do"
do "$root/day5/modules/05_report.do"
timer off 1
timer list 1

log close
```

TRAP

Never write `/*` in a comment line.

A header line such as `* Needs: day5/modules/*.do` contains `/*`: Stata opens a block comment there, and every line after it becomes a comment. The file runs, does nothing, and shows no error.

Write file lists in words: `the five files in day5/modules.`

YOUR TURN · TASK 25

Header and settings

DO

File > New > Do-file. Save it as `day5/master.do`. Type a header (Day 1 template), then:

```
version 17
clear all
global root "C:/NBS_Stata"
global quarter "2026Q1"
global youth_max 24
```

YOU SHOULD SEE

Nothing yet: settings are silent.

IF NOT

YOUR TURN · TASK 26

Check the inputs

DO

Add:

```
foreach f in "data/lfs_raw.dta"
"data/cpi_divisions_long.dta" {
    capture confirm file "$root/`f'"
    if _rc {
        display as error "Input missing:
$root/`f'"
        exit 601
    }
}
```

EXTRA Here `in` is right: two words, taken as written.

YOU SHOULD SEE

Silent: both files are there.

IF NOT

A folder, if it is missing:



YOUR TURN · TASK 27

What mkdir says

DO

Type:

```
capture mkdir "output"
display _rc
capture mkdir "output/d5_test"
display _rc
```

```
rmdir "output/d5_test"
```

YOU SHOULD SEE

693, then 0.

IF NOT

Two return codes

```
. capture mkdir "output"

. display _rc
693

. capture mkdir "output/d5_test"

. display _rc
0

. rmdir "output/d5_test"

.
. * ============================ PART 5  RELEASE ========================
```

1. 693: the folder already exists, harmless under `capture`; 0: made

One log for the whole run

```
capture log close
log using "$root/logs/master_${quarter}.log", replace text
display "Run on `c(current_date)' at Stata `c(stata_version)', quarter
$quarter"
```

- close any log a crash left open
- one log per quarter, named with the quarter
- the first line of the log says when, with what, for which quarter

COMMON ERROR

A log that is already open

```
. log using "logs/d5_tmp.log", replace text
log file already open
```

WHY

A run that stopped half-way leaves its log open. The next `log using` refuses.

FIX

`capture log close` on the line before.

YOUR TURN · TASK 28

Folders, log, modules

DO

Add, then save and run the whole file (Ctrl + D):

```
capture mkdir "$root/output"
capture mkdir "$root/logs"
capture log close
log using "$root/logs/master_${quarter}.log",
replace text
display "Run on `c(current_date)' at Stata
`c(stata_version)', quarter $quarter"
timer clear
timer on 1
do "$root/day5/modules/01_clean.do"
... the same for 02, 03, 04, 05 ...
timer off 1
timer list 1

log close
```

YOU SHOULD SEE

Five modules scroll past; the last line before the log closes is the timer.

IF NOT

Close Word and Excel first: modules 03 to 05 write to them.

The run starts

```
. display "Run on `c(current_date)' at Stata `c(stata_version)', quarter $quarter"
Run on 20 Oct 2026 at Stata 19.5, quarter 2026Q1
. * ---- The modules, in order -------------------------------------------
. timer clear

. timer on 1

. do "$root/day5/modules/01_clean.do"

. * ====================================================================
. * 01_clean.do  —  module 1: raw LFS file to the analysis file (Day 1)
. * --------------------------------------------------------------------
. * Course     : NBS Seychelles Stata training, Day 5 (20 Nov 2026)
```

1. the stamp: date, version, quarter
2. module 01 starts, header first

The run ends without red

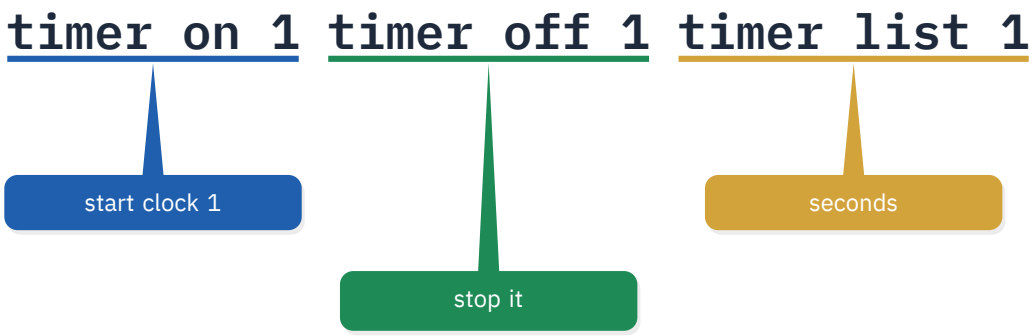
```
.
. putdocx save "$root/output/release_${quarter}.docx", replace
successfully replaced "C:/NBS_Stata/output/release_2026Q1.docx"
  1
.
end of do-file 2

. timer off 1

. timer list 1 3
   1:      8.24 /        1 =       8.2430
```

1. the release is written
2. module 05 ends
3. 8.24 seconds for the whole quarter

How long does it take?



YOUR TURN · TASK 29

One clock per module

DO

Type:

```
timer clear
timer on 1
quietly do "$root/day5/modules/01_clean.do"
timer off 1
... clocks 2 to 5 for modules 02 to 05 ...
timer list
```

YOU SHOULD SEE

Five lines, one per clock, in seconds.

IF NOT

EXTRA Lab 1 Challenge: the same in a loop.

Where the time goes

```
1:      0.09 /      1 =      0.0890
2:      0.02 /      1 =      0.0240
3:      0.79 /      1 =      0.7920
4:      4.82 /      1 =      4.8170
5:      0.45 /      1 =      0.4520
```

1

1 module 04 (the logit and two charts): 4.8 of about 6 seconds

A new colleague can now reproduce the quarter: change one line, press one key.

ON YOUR OWN · G

Next quarter

THE QUESTION

In April the 2026Q2 raw file arrives. Which line of `master.do` changes? What else must change?

HINT

Look at what module 01 opens.

ON YOUR OWN · G · ANSWER

```
global quarter "2026Q2"
* and module 01 reads lfs_raw.dta, a name with no quarter:
* the new raw file must replace it, or the name must change
```

```
use "$root/data/lfs_raw.dta", clear
```

1. the quarter names every output; the input name has no quarter in it. Part 7 fixes that.

QUICK CHECK

Why does master.do start with `capture log close`?

A. to save the log

B. a log left open by a crash would stop log using

C. to clear the data

ANSWER

Why does master.do start with `capture log close`?

A to save the log

B a log left open by a crash would stop log using

C to clear the data

Log S05: log file already open.

Part 4 on one page

CHEAT CARD · KEEP THIS ONE

`version 17 clear all`	first lines
`global root / quarter / ...`	the settings block
`foreach f in "a" "b" { capture confirm file ... }`	check inputs
`capture mkdir "$root/output"`	folders
`capture log close`	then log using, a log named with the quarter
`do "$root/day5/modules/01_clean.do"`	one line per module
`timer on 1 / timer off 1 / timer list`	how long

Practice on your own

PRACTICE SET · practice_D5.do, PART 4

CORE	P1	Regions sheet: type the rate of Central 1.
CORE	P2	Confirm the four products of the master exist.
CORE	P3	`capture mkdir` the output folder: type the code.
EXTENSION	P4	Margins sheet: type the predicted rate for 55+.
CHALLENGE	P5	CPI sheet: type the inflation figure; assert the month.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You have built a master do-file for a statistical product.

lost? open **catchup_D5_S4.do**

Lab 1 · Assemble the pipeline

THE SITUATION

Next quarter a colleague produces the release while you are on leave. She changes one block of settings, presses Ctrl + D, and gets today's files.

OPEN

`lab_D5_1.do` in `C:\NBS_Stata\day5\labs`. Close Word and Excel before each run.

THE TASKS

CORE Tasks 1 to 5: settings, input checks, folders and log, the five modules timed, the result read back

EXTENSION Task 6: one person, one row, or stop

CHALLENGE Task 7: one timer per module, in a loop

CORRECT WHEN

the whole file runs with no red text and ends with `end of do-file`.

The lab file: lab_D5_1.do

```
* TASK 2 (Core)  Check the inputs before anything runs
*   A foreach loop over the two input files (data/lfs_raw.dta and
*   data/cpi_divisions_long.dta). For each: capture confirm file; if it
*   fails, display a red message naming the file and exit 601.
* ------------------------------------------------------------------------
* >>> YOUR CODE


* CHECK Task 2 (silent = correct)
confirm file "$root/data/cpi_divisions_long.dta"
```

Type under >>> YOUR CODE. The CHECK lines are written for you.

Green note when your master runs from top to bottom without red.



5

DAY 5 · PART 5

Written by the do-file

THE PROBLEM

The figures are in Excel. The release is in Word. Someone copies them across.

WHAT WE HAVE

An Excel file of estimates and two charts, all written by the master.

WHAT WE NEED

The release text and tables, with the same numbers, written by the same run.

WHY NOT COPY

A copied number does not change when the data change. A typo in it is published.

The do-file writes the Word document too.

Part 5: the automated release

AFTER THIS PART YOU CAN

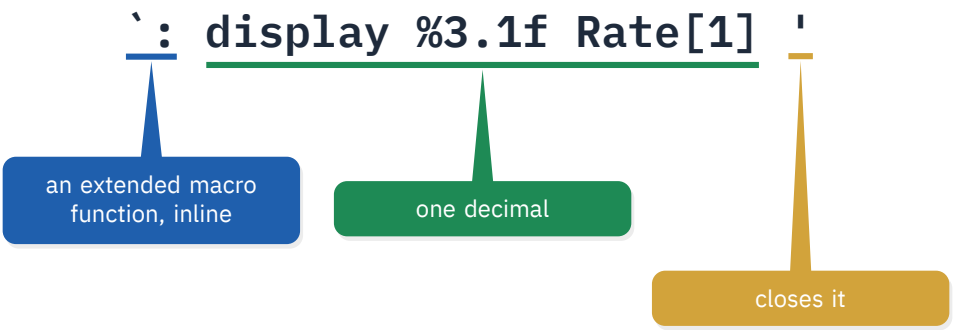
- turn a number into text with the format you choose
- write a sentence whose numbers come from the results
- fill a Word table in a loop
- rebuild the whole release after changing one setting

05_report.do opens the document

```
putdocx clear
putdocx begin
putdocx paragraph, style(Title)
putdocx text ("Labour Force Survey $quarter: key figures")
putdocx paragraph
putdocx text ("National Bureau of Statistics, Seychelles. Produced on
`c(current_date)'.")
```

Day 2, Part 8 commands. New: the quarter and the date come from macros.

A number, as text, with a format:



YOUR TURN · TASK 30

Three ways to show one rate

DO

Type:

```
import excel using
"$root/output/estimates_${quarter}.xlsx", ///
    sheet("National") firstrow clear
display "`=Rate[1]'"
display "[`: display %3.1f Rate[1]']"
display "[`: display %4.1f Rate[1]']"
```

YOU SHOULD SEE

Three versions of the unemployment rate.

IF NOT

EXTRA Which one belongs in a sentence?

Read the three versions

```
. display "`=Rate[1]'" 1
3.306604607329804

. display "[`: display %3.1f Rate[1]']" 2
[3.3]

. display "[`: display %4.1f Rate[1]']" 3
[ 3.3]

. display "[`: display %5.2f Rate[1]']"
[ 3.31]
```

1. all digits: no reader wants them
2. `%3.1f`: 3.3, right for a sentence
3. `%4.1f`: a space in front, so the sentence reads ”was 3.3”

A sentence written from the results

```
putdocx paragraph
putdocx text ("The unemployment rate was `: display %3.1f Rate[1]' per cent ")
putdocx text ("(95% interval `: display %3.1f Lower[1]' to `: display %3.1f Upper[1]').")
```

- the National sheet is in memory: row 1 is unemployment
- three numbers, none typed
- two `putdocx text` lines join into one paragraph

YOUR TURN · TASK 31

Title and date

DO

In `my_day5.do`, after the last task, type:

```
putdocx clear
putdocx begin
putdocx paragraph, style(Title)
putdocx text ("Labour Force Survey $quarter: key
figures")
putdocx paragraph
putdocx text ("Produced on `c(current_date)'.")
```

YOU SHOULD SEE

Silent: the document is in memory until it is saved.

IF NOT

YOUR TURN · TASK 32

The sentence

DO

Add:

```
putdocx paragraph
putdocx text ("The unemployment rate was `:
display %3.1f Rate[1]' per cent ")
putdocx text ("(95% interval `: display %3.1f
Lower[1]' to `: display %3.1f Upper[1]').")
```

YOU SHOULD SEE

Silent.

IF NOT

`variable Rate not found`? Run the `import excel` line of the task before.

One Word row per Excel row

```
putdocx table reg = (8, 4)
putdocx table reg(1, 1) = ("Region")
forvalues i = 1/7 {
local row = `i' + 1
putdocx table reg(`row', 1) = (Region[`i'])
putdocx table reg(`row', 2) = (Rate[`i']), nformat(%3.1f)
}
```

- 8 rows: a header and seven regions; 4 columns
- row `i + 1`: row 1 is the header
- `nformat(%3.1f)`: one decimal, as on Day 4

YOUR TURN · TASK 33

The regional table

DO

Type:

```
import excel using
"$root/output/estimates_${quarter}.xlsx", ///
    sheet("Regions") firstrow clear
putdocx table reg = (8, 4)
putdocx table reg(1, 1) = ("Region")
... reg(1, 2) Rate, reg(1, 3) Lower, reg(1, 4)
Upper ...
forvalues i = 1/7 {
    local row = `i' + 1
    putdocx table reg(`row', 1) = (Region[`i'])
    putdocx table reg(`row', 2) = (Rate[`i']),
nformat(%3.1f)
    ... columns 3 and 4: Lower, Upper ...
}
```

YOU SHOULD SEE

Silent.

IF NOT

COMMON ERROR

A cell outside the table

```
. putdocx table t = (2, 2)
. putdocx table t(3, 1) = ("North")
invalid cell specification; row out of range
```

WHY

The table has 2 rows; row 3 does not exist. With a loop, the table size and the loop must agree.

FIX

Rows = header + regions: `(8, 4)` for 7 regions.

YOUR TURN · TASK 34

The chart, then save

DO

Type:

```
putdocx paragraph
putdocx image
"$root/output/margins_${quarter}.png"
putdocx save
"$root/output/my_release_${quarter}.docx", replace
```

YOU SHOULD SEE

One line: `successfully created`. Open the file.

IF NOT

File in use? Close it in Word, run again.

Your release is written

```
. putdocx save "output/my_release_2026Q1.docx", replace
successfully created "C:/NBS_Stata/output/my_release_2026Q1.docx"
```

1. the file name carries the quarter

COMMON ERROR

Text before the document is open

```
. putdocx text ("The rate was 3.3 per cent.")
document not opened
```

WHY

`putdocx clear` closed the document. Text needs an open document.

FIX

`putdocx begin` first, then paragraphs and text.

The release interval: why proportion

MEAN: LINEARISED INTERVAL

```
             |       Mean   std. err.     [95% conf. interval]
-------------+------------------------------------------------
       unemp |   .0083458   .0042431     -.0006492    .0173408
```

1

PROPORTION: LOGIT INTERVAL

```
             | Proportion   std. err.     [95% conf. interval]
-------------+------------------------------------------------
       unemp |
          0  |   .9916542   .0042431      .9756543    .9971695
          1  |   .0083458   .0042431      .0028305    .0243457
```

2

1 Praslin & La Digue: lower limit −.0006, a negative rate 2 the same 0.83%, from .28% to 2.43%. As on Wednesday: the release uses `proportion`.

Which column holds the rate?

```
. matrix list r(table), format(%9.5f)

r(table)[9,2]
                 0.          1.
              unemp       unemp
     b     0.99165     0.00835
    se     0.00424     0.00424
     t   233.70802     1.96689
pvalue     0.00000     0.06679
    ll     0.97565     0.00283
    ul     0.99717     0.02435
```

1. column 1 is level 0 (not unemployed), column 2 level 1: module 03 reads `T[1, 2]`, `T[5, 2]`, `T[6, 2]`

YOUR TURN · TASK 35

Run the master

DO

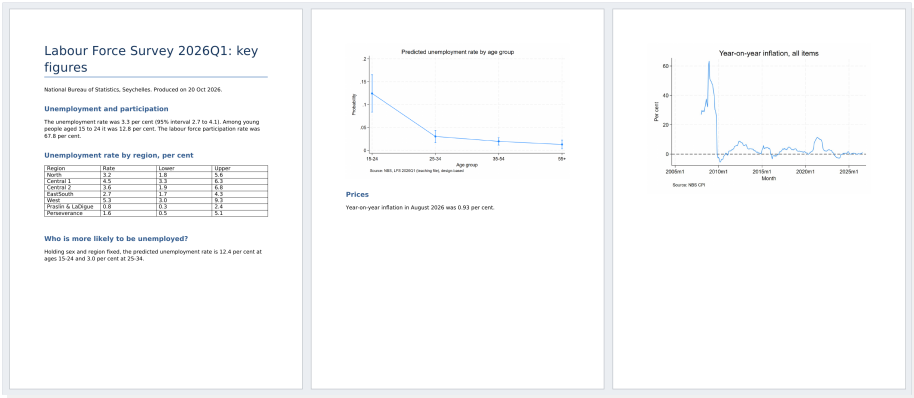
Close Word. Open `master.do` and press Ctrl + D.

YOU SHOULD SEE

The full run, then `output/release_2026Q1.docx`: three pages.

IF NOT

The finished mini-release



Labour Force Survey 2026Q1: key figures

National Bureau of Statistics, Seychelles. Produced on 20 Oct 2026.

Unemployment and participation

The unemployment rate was 3.3 per cent (95% interval 2.7 to 4.1). Among young people aged 15 to 24 it was 12.8 per cent. The labour force participation rate was 67.8 per cent.

Unemployment rate by region, per cent

Region	Rate	Lower	Upper
North	3.2	1.8	5.6
Central 1	4.5	3.3	6.3
Central 2	3.6	1.9	6.8
EastSouth	2.7	1.7	4.3
West	5.3	3.0	9.3
Praslin & LaDigue	0.8	0.3	2.4
Perseverance	1.6	0.5	5.1

Who is more likely to be unemployed?

Holding sex and region fixed, the predicted unemployment rate is 12.4 per cent at ages 15-24 and 3.0 per cent at 25-34.

Prices

Year-on-year inflation in August 2026 was 0.93 per cent.

output/release_2026Q1.docx, three pages, written by `master.do` in one run.

Read page 1



Labour Force Survey 2026Q1: key figures

National Bureau of Statistics, Seychelles. Produced on 20 Oct 2026.

Unemployment and participation

The unemployment rate was 3.3 per cent (95% interval 2.7 to 4.1). Among y people aged 15 to 24 it was 12.8 per cent. The labour force participation rate 67.8 per cent.

Unemployment rate by region, per cent

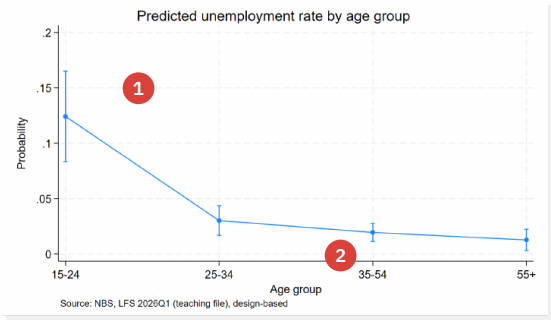
Region	Rate	Lower	Upper
North	3.2	1.8	5.6
Central 1	4.5	3.3	6.3
Central 2	3.6	1.9	6.8
EastSouth	2.7	1.7	4.3
West	5.3	3.0	9.3
Praslin & LaDigue	0.8	0.3	2.4
Perseverance	1.6	0.5	5.1

Who is more likely to be unemployed?

Holding sex and region fixed, the predicted unemployment rate is 12.4 per cent at ages 15-24 and 3.0 per cent at 25-34.

1. quarter from $quarter, date from c()
2. three figures from the National sheet
3. the table: seven rows from the loop
4. Praslin: 0.3 to 2.4, no negative rate

Read page 2



1. the Day 4 design-based margins chart
2. source line from the do-file

Every number in this document came from the data in this run.
None was typed.

YOUR TURN · TASK 36

The proof

DO

In `master.do` change one setting, then run:

```
global youth_max 29
```

YOU SHOULD SEE

The whole release again, in seconds. The youth sentence now says 15 to 29.

IF NOT

The estimates change everywhere

```
    +-----------------------------------------------+
    | Indica~r        Rate       Lower       Upper |
    |-----------------------------------------------|
1.  |    unemp   3.3066046   2.6831721   4.0688347 |
2.  |    youth   8.2771773   6.4001942   10.642037 |
3.  |     part   67.807314   65.811933   69.739783 |
    +-----------------------------------------------+
```

1

1 youth (15-29): 8.3%, from 6.4 to 10.6. Unemployment and participation do not move

The same sentence, two definitions

YOUTH 15-24

Unemployment and participation

The unemployment rate was 3.3 per cent (95% interval 2.7 to 4.1). Among young people aged 15 to 24 it was 12.8 per cent. The labour force participation rate was 67.8 per cent.

YOUTH 15-29

Unemployment and participation

The unemployment rate was 3.3 per cent (95% interval 2.7 to 4.1). Among young people aged 15 to 29 it was 8.3 per cent. The labour force participation rate was 67.8 per cent.

One line changed in the settings block; the text follows.

YOUR TURN · TASK 37

Undo

DO

Set it back and run again:

```
global youth_max 24
```

YOU SHOULD SEE

The original release returns: youth 12.8%.

IF NOT

Back to 15-24

```
    +-----------------------------------------------+
    | Indica~r        Rate       Lower       Upper |
    |-----------------------------------------------|
1.  |    unemp   3.3066046   2.6831721   4.0688347 |
2.  |    youth   12.776274   9.1196679   17.614851 |
3.  |     part   67.807314   65.811933   69.739783 |
    +-----------------------------------------------+
```

1

1. 12.8%, as before: the run is reproducible

ON YOUR OWN · I

One more sentence

THE QUESTION

Write the sentence for participation: rate and 95% interval, one decimal, from the National sheet.

HINT

Row 3.

ON YOUR OWN · I · ANSWER

```
display "The participation rate was `: display %3.1f
Rate[3]' per cent ///
(95% interval ...)."
```

```
The participation rate was 67.8 per cent (95% interval 65.8 to 69.7).
```
(1)

(1) the sentence, ready for `putdocx text`

ON YOUR OWN · J

The least precise region

THE QUESTION

Which region has the widest interval in the release table? Why does that matter for the ranking?

HINT

Upper minus lower.

ON YOUR OWN · J · ANSWER

```
import excel using ///
"$root/output/estimates_${quarter}.xlsx", ///
sheet("Regions") firstrow clear
list
```

```
     +-------------------------------------------------------+
     |            Region        Rate       Lower       Upper |
     |-------------------------------------------------------|
  1. |             North   3.2063818   1.8149003   5.6038207 |
  2. |         Central 1   4.5333616   3.2704103   6.2525075 |
  3. |         Central 2   3.6178327   1.8938999    6.802183 |
  4. |         EastSouth   2.6792267    1.666633   4.2802614 |
  5. |              West   5.3162335   2.9868997    9.288181 |
     |-------------------------------------------------------|
  6. | Praslin & LaDigue    .8345796    .28304638   2.4345706 |
  7. |      Perseverance   1.6284564    .50653157   5.1077772 |
     +-------------------------------------------------------+
```

1

1 West: 3.0 to 9.3, the highest rate and the widest interval. Its first place is not firm

QUICK CHECK

The release shows `was  3.3`. Which format caused the double space?

A. %3.1f

B. %4.1f

C. %5.2f

ANSWER

The release shows `was  3.3`. Which format caused the double space?

A %3.1f

B %4.1f

C %5.2f

Log R01: %4.1f pads 3.3 to four characters.

Part 5 on one page

CHEAT CARD · KEEP THIS ONE

`` `: display %3.1f x' ``	a number as text, one decimal
`putdocx text ("... `: display %3.1f Rate[1]' ...")`	a sentence from the results
`putdocx table t = (8, 4)`	a table
`putdocx table t(r, c) = (x), nformat(%3.1f)`	one cell
`putdocx image "file.png"`	a chart
`putdocx save "f_${quarter}.docx", replace`	write it
`svy, subpop(): proportion x, citype(logit)`	release intervals

Practice on your own

PRACTICE SET · practice_D5.do, PART 5

CORE	P1	The participation rate as text, one decimal.
CORE	P2	A Word file with a title and the unemployment sentence.
CORE	P3	A Word table of the three national figures.
EXTENSION	P4	A Word file with the inflation chart and its sentence.
CHALLENGE	P5	Youth 15-29 unemployment, without changing any file.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

One command rebuilds NBS’s Q1 2026 mini-release. That is today’s product.

lost? open **catchup_D5_S5.do**



6

DAY 5 · PART 6

Making an old do-file safe

THE PROBLEM

A do-file that has produced the ILO table for years stops working on a new laptop

WHAT WE HAVE

A short file, edited by several people, that runs on one machine.

WHAT WE NEED

The same results, on any machine, with a log and a clear stop if the data are missing.

WHY IT BREAKS

Paths to a drive only one office has, settings switched off, no version, no log.

Three fixes make it reproducible. The numbers do not change.

Part 6: refactoring

AFTER THIS PART YOU CAN

- read an inherited do-file and find what ties it to one machine
- replace absolute paths by paths from `$root`
- gather the settings into one block at the top
- add a version, a header, a log and an input check

Most production do-files grow over years. Here is how to make one reproducible.

The case file: legacy_ilo.do

```
* ILO indicators LFS - quarterly
* updated many times, last by JP
*cd "Z:\LFS\2026\Q1\"
*global wgt weights_dis
*global qtr "Q1"

use "Z:\LFS\2026\Q1\data\lfs_raw.dta", clear
drop if ilo_age<15
gen lf=1 if ilo_lfs19==1|ilo_lfs19==2
replace lf=0 if lf==.
gen un=ilo_lfs19==2
* unemployment rate
tab un if lf==1 [iw=$wgt]
* by sex
tab un dem_sex if lf==1 [iw=$wgt], col
* by region
tab un region_q if lf==1 [iw=$wgt], col nofreq
save "Z:\LFS\2026\Q1\data\lfs_ilo_$qtr.dta", replace
```

Written for this course with the problems we meet in real files. Read it: what will happen on your laptop?

Problem 1: a drive only one office has

```
* ILO indicators LFS - quarterly
* updated many times, last by JP
*cd "Z:\LFS\2026\Q1\"
*global wgt weights_dis
*global qtr "Q1"

use "Z:\LFS\2026\Q1\data\lfs_raw.dta", clear
drop if ilo_age<15
gen lf=1 if ilo_lfs19==1|ilo_lfs19==2
replace lf=0 if lf==.
gen un=ilo_lfs19==2
* unemployment rate
tab un if lf==1 [iw=$wgt]
* by sex
tab un dem_sex if lf==1 [iw=$wgt], col
* by region
tab un region_q if lf==1 [iw=$wgt], col nofreq
save "Z:\LFS\2026\Q1\data\lfs_ilo_$qtr.dta", replace
```

Z: is a network drive in one office. On a laptop it does not exist.

Problem 2: settings switched off

```
* ILO indicators LFS - quarterly
* updated many times, last by JP
*cd "Z:\LFS\2026\Q1\"
*global wgt weights_dis
*global qtr "Q1"

use "Z:\LFS\2026\Q1\data\lfs_raw.dta", clear
drop if ilo_age<15
gen lf=1 if ilo_lfs19==1|ilo_lfs19==2
replace lf=0 if lf==.
gen un=ilo_lfs19==2
* unemployment rate
tab un if lf==1 [iw=$wgt]
* by sex
tab un dem_sex if lf==1 [iw=$wgt], col
* by region
tab un region_q if lf==1 [iw=$wgt], col nofreq
save "Z:\LFS\2026\Q1\data\lfs_ilo_$qtr.dta", replace
```

The globals are commented out, yet used below. They work only if someone set them by hand earlier in the session.

Problem 3: no version, no log, no check

```
* ILO indicators LFS - quarterly
* updated many times, last by JP
*cd "Z:\LFS\2026\Q1\"
*global wgt weights_dis
*global qtr "Q1"

use "Z:\LFS\2026\Q1\data\lfs_raw.dta", clear
drop if ilo_age<15
gen lf=1 if ilo_lfs19==1|ilo_lfs19==2
replace lf=0 if lf==.
gen un=ilo_lfs19==2
* unemployment rate
tab un if lf==1 [iw=$wgt]
* by sex
tab un dem_sex if lf==1 [iw=$wgt], col
* by region
tab un region_q if lf==1 [iw=$wgt], col nofreq
save "Z:\LFS\2026\Q1\data\lfs_ilo_$qtr.dta", replace
```

The header says who touched it last, not what it needs or makes. No `version`, no `log`, no `confirm file`.

YOUR TURN · TASK 38

Run it as found

DO

Copy `day5/refactor/legacy_ilo.do` to `day5/refactor/my_ilo.do`. Open the copy and run it.

YOU SHOULD SEE

Red text at the `use` line.

IF NOT

EXTRA Keep the original: you will compare.

It stops at the first path

```
.
. use "Z:\LFS\2026\Q1\data\lfs_raw.dta", clear
file Z:\LFS\2026\Q1\data\lfs_raw.dta not found 1
r(601);
```

1 the Z: drive is not on this laptop: r(601)

COMMON ERROR

The next error, hiding behind the first

```
. macro drop wgt qtr
. tabulate ilo_sex [iweight = $wgt]
invalid syntax
```

WHY

With the path fixed, the `tab` lines meet `$wgt`. It was never set: `[iweight = ]` is not valid syntax.

FIX

Fix 2: set every global the file uses, at the top.

YOUR TURN · TASK 39

Fix 1: paths from the root

DO

In `my_ilo.do` replace the two Z: paths:

```
use "$root/data/lfs_raw.dta", clear
save "$root/data/lfs_ilo_${quarter}.dta", replace
```

YOU SHOULD SEE

Not yet: the globals are not set. Fix 2 next.

IF NOT

YOUR TURN · TASK 40

Fix 2: one settings block

DO

At the top, after the header, type:

```
version 17
clear all
global root "C:/NBS_Stata"
global quarter "2026Q1"
global wgt "weights_dis"
```

YOU SHOULD SEE

Delete the three commented lines: one place for settings.

IF NOT

EXTRA Name: `quarter`, as in master.do, not `qtr`.

YOUR TURN · TASK 41

Fix 3: a header, a log, a check

DO

Add a header (Day 1 template), then after the settings:

```
capture log close
log using "$root/logs/my_ilo.log", replace text
capture confirm file "$root/data/lfs_raw.dta"
if _rc {
    display as error "Input missing:
$root/data/lfs_raw.dta"
    exit 601
}
 ... the original lines, with duplicates drop
added ...
log close
```

YOU SHOULD SEE

Run: three tables, a saved file, no red.

IF NOT

The refactored file runs on a laptop

```
. use "$root/data/lfs_raw.dta", clear
(NBS LFS Q1 2026 - workshop teaching extract (raw))
. tabulate un if lf == 1 [iweight = $wgt]

         un |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 | 52,431.644       96.69       96.69
          1 | 1,792.9944        3.31      100.00
------------+-----------------------------------
      Total | 54,224.639      100.00
```

1 the path from the root opens the file

2 3.31% unemployed: the weighted share

The other two tables

```
           |          Sex
        un |      MALE     FEMALE |     Total
-----------+----------------------+----------
         0 | 25,059.95  27,371.69 | 52,431.64
           |     96.55      96.83 |     96.69
-----------+----------------------+----------
         1 | 896.80564  896.18876 |1,792.9944
           |      3.45       3.17 |      3.31
           |                                                                  Region
        un |  1- North  2- Centra  3- Centra  4- EastSo    5- West  6- Prasli  7- Persev |     Total
-----------+-------------------------------------------------------------------------------+----------
         0 |     96.79      95.47      96.38      97.32      94.68      99.17      98.37 |     96.69
         1 |      3.21       4.53       3.62       2.68       5.32       0.83       1.63 |      3.31
```

1. men 3.45%, women 3.17%

2. the seven regions: the same rates as the loop in Part 2

Before and after

BEFORE

```
*cd "Z:\LFS\2026\Q1\"
*global wgt weights_dis
*global qtr "Q1"

use "Z:\LFS\2026\Q1\data\lfs_raw.dta",
clear
```

AFTER

```
* ---- Fix 2: one settings block; every global set once
--------------
version 17
clear all
global root "C:/NBS_Stata"
global quarter "2026Q1"
global wgt "weights_dis"
```

The analysis lines barely change. What changes is everything around them: where the data are, what the file needs, what it leaves behind.

NOTE

The rate is the same. The interval is missing.

`tabulate` with `iweight` gives 3.31%, the same point estimate as the release, but no standard error.

For publication: `svy:` as in module 03.

ON YOUR OWN · K

Two lines into one

THE QUESTION

The legacy file builds `lf` with `gen ... if` and then `replace lf=0 if lf==.`. Write it in one line.

HINT

A condition in brackets is 1 or 0.

ON YOUR OWN · K · ANSWER

```
generate lf = (ilo_lfs19 == 1 | ilo_lfs19 == 2)
```

```
. generate lf = (ilo_lfs19 == 1 | ilo_lfs19 == 2)
. tabulate un if lf == 1 [iweight = $wgt]

         un |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 | 52,431.644       96.69       96.69
          1 | 1,792.9944        3.31      100.00
------------+-----------------------------------
      Total | 54,224.639      100.00
```

1

1 the same table as the legacy definition gives

ON YOUR OWN · L

Same number, different status

THE QUESTION

The refactored file and the release both give 3.3%. Which one may NBS publish, and why?

HINT

Compare F03 with the National sheet.

ON YOUR OWN · L · ANSWER

```
* refactored file: tabulate ... [iweight = $wgt]
* release: svy, subpop(lf): proportion unemp,
citype(logit)
```

```
     +-----------------------------------------------+
     | Indica~r        Rate       Lower       Upper |
     |-----------------------------------------------|
  1. |    unemp   3.3066046   2.6831721   4.0688347 |  (1)
     +-----------------------------------------------+
```

(1) the release: 3.3% with its interval 2.7 to 4.1, from the declared design

QUICK CHECK

Which line ties the legacy file to one office?

A. use ”Z:\LFS\2026\Q1\data\lfs_raw.dta”

B. gen un=ilo_lfs19==2

C. tab un if lf==1

ANSWER

Which line ties the legacy file to one office?

A use ”Z:\LFS\2026\Q1\data\lfs_raw.dta”

B gen un=ilo_lfs19==2

C tab un if lf==1

Log F01: r(601) at the Z: path.

Part 6 on one page

CHEAT CARD · KEEP THIS ONE

`Z:\... C:\Users\...`	absolute path: replace by the root global
`* global ... (commented)`	settings switched off: one block at the top
`$name used, never set`	set every global the file uses
`version 17`	first command
`log using "$root/logs/x.log"`	a record of the run
`capture confirm file ... exit 601`	stop with a message

Practice on your own

PRACTICE SET · practice_D5.do, PART 6

CORE	P1	Weighted unemployment rate, the legacy way, from the raw file.
CORE	P2	The same rate for women.
CORE	P3	The same rate for Praslin & La Digue.
EXTENSION	P4	How many persons does `drop if ilo_age < 15` remove?
CHALLENGE	P5	Assert that the one-line and the two-line `lf` agree.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You can take any inherited do-file and make it run on any machine.

lost? open **catchup_D5_S6.do**

7

DAY 5 · PART 7

Habits that protect it

THE PROBLEM

A release is right today. Will it still be right in a year?

WHAT WE HAVE

A master, five modules, a release, built this week.

WHAT WE NEED

The same release next year, on another Stata, by someone else.

WHAT BREAKS IT

An overwritten raw file, a hand-edited table, a new Stata, three copies of the master.

Five habits, each one line or one rule.

Part 7: discipline

AFTER THIS PART YOU CAN

- keep raw data read-only and outputs rebuilt by the run
- name files so that they sort and tell their story
- pin the Stata version in every file
- notice when a dataset has changed

Three kinds of file

```
C:/NBS_Stata/
    data/                          raw files: read, never overwritten
        lfs_raw.dta                from the LFS team
        lfs_2026Q1_teach.dta       made by module 01: can be rebuilt
    day5/modules/, master.do       the only files edited by hand
    output/                        regenerated by every run
    logs/                          the record of every run
```

Never overwrite raw data.

Never edit an output by hand.

Fix the do-file and rerun.

YOUR TURN · TASK 42

Stata protects raw data

DO

Type:

```
use "$root/data/lfs_raw.dta", clear
capture save "$root/data/lfs_raw.dta"
display _rc
```

YOU SHOULD SEE

602: nothing was written.

IF NOT

COMMON ERROR

save without replace

```
. save "data/lfs_raw.dta"
file data/lfs_raw.dta already exists
```

WHY

The file exists, so `save` refuses. That is protection, not a fault.

FIX

Never add `replace` to a save of a raw file. Save under a new name.

YOUR TURN · TASK 43

File names that tell the story

DO

Type:

```
local files : dir "$root/output" files "*2026Q1*"
foreach f of local files {
    display "output: `f'"
}
```

YOU SHOULD SEE

The five outputs of the master.

IF NOT

Five names, one pattern

```
output: estimates_2026q1.xlsx (1)
output: infl_2026q1.png
output: margins_2026q1.png
output: model_2026q1.xlsx
output: release_2026q1.docx (2)
```

1. what, then the quarter, then the type
2. Windows shows the names in small letters; Stata still finds them

The first line of every do-file:

```
version 17
```

run as Stata 17 would,
on any later Stata

YOUR TURN · TASK 44

Which version is running?

DO

Type:

```
version
display c(stata_version)
```

YOU SHOULD SEE

Two numbers: the version the file asks for, the version installed.

IF NOT

Asked for, and installed

```
. version
version 17.0 (1)

. display c(stata_version)
19.5
```

(1) the do-file behaves as version 17, on a newer Stata

Has the file changed?

`datasignature set`

record a fingerprint of the data

`datasignature confirm`

compare: 0 = same

YOUR TURN · TASK 45

A fingerprint

DO

Type:

```
use "$root/data/lfs_${quarter}_svy.dta", clear
datasignature set, reset
capture datasignature confirm
display _rc
replace ilo_age = ilo_age + 1 in 1
capture datasignature confirm
display _rc
```

YOU SHOULD SEE

The fingerprint, 0, one change, then 9.

IF NOT

One change in 5,888 rows is noticed

```
. datasignature set, reset
  5888:93(83996):663834272:4023702331       (data signature set)

. capture datasignature confirm

. display _rc
0

. replace ilo_age = ilo_age + 1 in 1
(1 real change made)

. capture datasignature confirm

. display _rc
9 1
```

1. the fingerprint, then 0: unchanged; after one edit, 9: changed

Next quarter

```
global quarter "2026Q2"
use "$root/data/lfs_${quarter}_raw.dta", clear
```

- change the quarter in the settings block
- better: name the raw file with its quarter too, and module 01 needs no edit
- then run master.do

Version control in one picture

DATED COPIES

```
master_final.do
master_final2.do
master_final_JP.do
master_USE_THIS.do
```

Which one made the March release?
Nobody knows.

A HISTORY OF CHANGES

```
one master.do
+ each change: who, when, why
+ any earlier version on request
```

Git, the tool behind this course's repository, keeps that history.

ON YOUR OWN · M

Was the file touched?

THE QUESTION

You receive `lfs_2026Q1_svy.dta` from a colleague. How do you know whether it is the file your run made?

HINT

`datasignature`.

ON YOUR OWN · M · ANSWER

```
datasignature set, reset
* ... later, on the copy you receive:
capture datasignature confirm
display _rc
```

```
. datasignature set, reset
  5888:93(83996):663834272:4023702331       (data signature set)

. capture datasignature confirm

. display _rc
0

. replace ilo_age = ilo_age + 1 in 1
(1 real change made)

. capture datasignature confirm

. display _rc
9 1
```

1. 0: the same data; 9: something changed

QUICK CHECK

`save "data/lfs_raw.dta"` without `replace` gives r(602). Should you add `replace`?

A. yes, the error must go

B. no: save under a new name; raw data are never overwritten

C. only if the file is small

ANSWER

`save "data/lfs_raw.dta"` without `replace` gives r(602). Should you add `replace`?

A. yes, the error must go

B. no: save under a new name; raw data are never overwritten

C. only if the file is small

Log V05.

Part 7 on one page

CHEAT CARD · KEEP THIS ONE

`data/ raw files`	read only; save without replace refuses
`output/, logs/`	rebuilt by every run
`what_quarter.type`	names that sort and tell
`version 17`	first command, every file
`datasignature set / confirm`	has the data changed?
`one master.do + history`	not five dated copies

Practice on your own

PRACTICE SET · practice_D5.do, PART 7

CORE	P1	Type the version the file runs under.
CORE	P2	`datasignature`: the code before a change.
CORE	P3	The code after a change.
EXTENSION	P4	Save the raw file without replace: the code.
CHALLENGE	P5	How many do-files in `day5/modules`? (`word count`)

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You leave with the habits that keep an NBS release reproducible quarter after quarter.

lost? open `catchup_D5_S7.do`

Daily check

1. Open `check_D5.do` from the day5 folder.
2. Type your **participant code** on the first line.
3. One sentence: what was unclear today?
4. Close Word and Excel. Save and run with **Ctrl + D**.

Read the last line: how many of the six products are in place. It is not a grade.

8

DAY 5 · PART 8

What the week has built

Fifteen questions, the same skills as Monday morning. This measures the week, not you.

Final assessment

1. Open the form: link or QR code on the screen.
2. Enter your **participant code** only, no name.
3. Fifteen questions. No Stata needed.
4. Wi-Fi down? Paper copies at the front.



`forms.gle/NZJditMtEAFd5aET6`

Results go to NBS as room totals, compared with Monday's diagnostic.

The applied exercise · Youth in the labour market

THE SITUATION

The Ministry of Youth asks for three figures on young people aged 15 to 24, Q1 2026, each with its 95% interval.

OPEN

`day5/final/final_applied.do`. Your notes, the cheat-sheets and Stata's help are allowed.

THE TASKS

TASK 1 settings, log, modules 01 and 02

TASKS 2-4 youth unemployment, participation, by sex

TASK 5 the four rates to Excel; close the log

CORRECT WHEN

every CHECK is silent; copy `logs/final_<code>.log` to the USB folder `final_logs`.

The exercise file

```
* TASK 2  Youth unemployment rate
*   Design-based unemployment rate of young people in the labour force,
*   with its 95% interval: proportion, as in module 03. Copy r(table)
*   into Y.
* ------------------------------------------------------------------------
* >>> YOUR CODE
```

Each task ends with a CHECK. Silent = correct.

COMMON ERROR

If your first run says this

```
. assert "`code'" != ""
assertion is false
r(9);
```

WHY

The participant code is still empty. The first CHECK tests it before anything else.

FIX

Type your code on the `local code` line, then run again.

Done? Green note. Copy your log to the USB folder `final_logs`.

Feedback

Eleven short questions, anonymous: no name, no code.

What helped, what did not, what NBS should teach next.



forms.gle/x1pKvG52YtnD7kDL9

The week in five products

1. Monday: a clean, labelled analysis file
2. Tuesday: a tables pack in Excel
3. Wednesday: official estimates with their intervals
4. Thursday: an analytical note with a model and a chart
5. Friday: one master do-file that rebuilds them all and writes the release

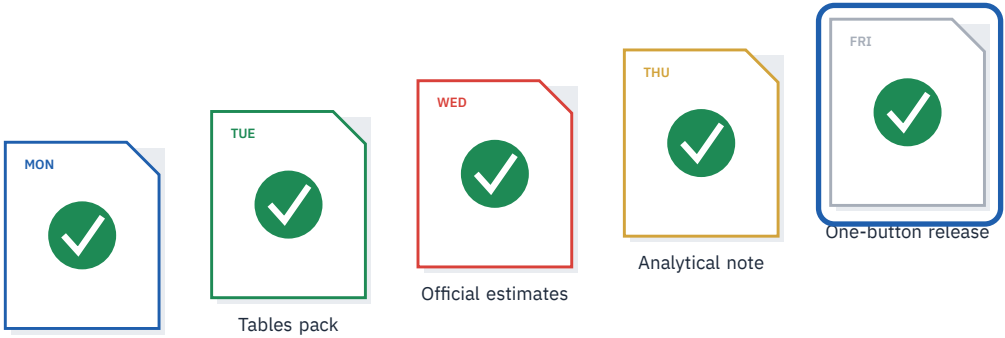
FROM

`lfs_raw.dta`

TO

`release_2026Q1.docx`, in one run

The completed stack



Where everything lives after today

- the course folder `C:\NBS_Stata`: data, do-files, outputs, logs
- the USB stick: the same folder, slides, workbook and cheat-sheets, offline packages
- the course repository `github.com/endri81/NBS-Stata-2026-participants`: every file, every update
- Stata's own help: `help` and the PDF manuals linked from it

These materials belong to NBS. Adapt them, teach them, improve them.

Certificates.

Thank you. Mersi bokou.